Power Basics

- **Power ($1 - \beta$):** Probability the test correctly rejects a false H_0.
- Depends on α, effect size, sample size, and population variability.
- Higher power means a lower chance of missing a real effect.

General Steps for Power Analysis

1. Specify H_0 and H_1 (including effect size of interest, e.g., difference Δ).
2. Choose α and tail direction.
3. Determine sample size (n) and distribution parameters.
4. Compute the noncentrality parameter or standardized effect.
5. Calculate β (probability of failing to reject) using statistical software or tables; power = $1 - \beta$.

Example 1 — One-Sample z-Test

- H_0: $\mu = 72$, H_1: $\mu \neq 72$, σ known = 10, $n = 36$, $\alpha = 0.05$.
- Critical $z \approx \pm 1.96$; margin = $1.96 \times (10/\sqrt{36}) \approx 3.27$.
- If true mean is 75 ($\Delta = 3$), z-shift = $\Delta / (\sigma/\sqrt{n}) = 3 / (10/6) = 1.8$.
- $\beta \approx P(|Z| < 1.96 - 1.8) \approx 0.36 \rightarrow$ Power ≈ 0.64.

Example 2 — Two-Sample t-Test (Equal Variances)

- H_0: $\mu_1 = \mu_2$, H_1: $\mu_1 \neq \mu_2$, σ assumed equal with $s = 12$, $n_1 = n_2 = 15$, $\alpha = 0.05$.
- df = 28, critical $t \approx \pm 2.048$.
- If true difference $\Delta = 8$, standard error = $s\sqrt{(2/n)} \approx 4.38$, noncentrality $\lambda = \Delta/SE \approx 1.83$.
- Approximate power via noncentral t or software (≈ 0.62). Increasing to $n = 25$ per group boosts power near 0.85.

Example 3 — Chi-Square Goodness-of-Fit

- H_0: observed proportions match expected; $\alpha = 0.05$, df = $k - 1$.
- Power depends on effect size $w = \sqrt{\Sigma((p\_obs - p\_exp)^2/p\_exp)}$.
- With $w = 0.3$ (medium) and df = 4, $n \approx 88$ yields power ≈ 0.80.

Notes on Nonparametric Tests

Closed-form power formulas may not exist; rely on simulation or reference tables. Document limitations in each topic’s notes.

Improving Power

- Increase sample size.
- Reduce measurement noise.
- Use one-tailed tests when justified (and pre-specified).
- Accept slightly higher α only if Type I consequences are tolerable.

Checklist

- Report assumed effect size and α when stating power.
- Mention how power influenced sample-size decisions.
- For the MA235 project, remind students that $n=15$ per group fixes power; interpreting results should note this limitation.